What's expected for submission:

1. A ~~1 minute, 30 second~~ 3 minute presentation (PPT, PDF, or similar) that covers
    1. Your problem space (approximately 40 seconds)
        - **Use a person's self-reported ratings (how attractive, fun, smart, sincere, ambitious they think they are) to predict their match rate.**
    2. Your dataset (approximately 80 seconds). This information should include, as applicable:

        1. The number of entries: 8,378
        2. The number of features per entry: 123
        3. A description of some of the most interesting or important features: Ambition, attractive, `sincere`, `intelligence, funny, expected_num_interested_in_me, expected_num_matches`
        4. The variable(s) you've chosen as your outcomes: Match

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3. A description of your planned approach (approximately 60 seconds), including
    - Your initial methods for implementing a solution
        i. We plan to begin by selecting the five self-reported personality ratings: attractiveness, funniness, intelligence, sincerity, and ambition. We'll clean the data by handling missing values and then use these variables to predict the binary match outcome. We'll start with logistic regression as an interpretable baseline, then compare it against a decision tree and random forest to see whether nonlinear relationships improve prediction. We'll evaluate the models using cross-validation and metrics.
    - (If applicable / if you have time) How your idea differs from / builds on ideas from the research field
    - Consider adding "Stretch Goals" to your project approach / goals. These would be above and beyond your typical expectations for completion, but extra directions you might like to pursue if you find you complete your initial work with time to spare.

        i. Step wise feature selection to find the features that are most predictive of a match within the dataset.
1. A short (4 pages or less, and 1.5 to 2.5 pages should be fine) write-up of your presentation, including all of the areas above, as a PDF or DOC / DOCX
    1. This paper will dive into more details than your presentation can cover. You should focus much of your energy on presenting the data in a reasonable way and outlining how your planned approach is different from other research in the field.
        1. Cite at least 2 other research papers or corporate implementations in the problem space and / or methods areas.

Project Proposal Write-Up:

Problem Space:

People routinely describe themselves as attractive, funny, intelligent, sincere, and ambitious, and dating platforms lean heavily on this kind of self-presentation. However, how much do these self-perceptions actually tell us about romantic outcomes? We ask a question: can a person's self-reported ratings on those five traits predict whether a date results in a mutual match?

This matters for two reasons. First, recommendation systems in dating often rely on self-descriptions because they are cheap to collect, so it is useful to know how much signal they carry. Second, the question connects to a long-running debate in social psychology about whether stated traits and preferences predict real-world attraction. We treat this as a binary classification problem where the outcome is match versus no match, and we are as interested in the size of the signal as in maximizing accuracy.

Dataset:

We use the Speed Dating dataset hosted on OpenML (ID 40536), derived from the speed-dating experiments of Fisman et al. (2006) at Columbia University. In each event, participants held a series of short dates, then recorded ratings of their partner and a yes/no decision on whether to see the partner again. A match occurs when both people say yes. The dataset contains 8,378 entries, each one a single date between a participant

and a partner, with 123 features per entry. Because participants date many partners, the same person appears in many rows, which affects how we must validate our models.

Key feature groups:

Feature	Description
attractive, funny, intelligence, sincere, ambition	Participant's self-rating (1–10) on each trait; our five core predictors
expected_num_interested_in_me, expected_num_matches	Participant's pre-event guess of how many partners will be interested / match with them
attractive_important, funny_important	How the participant allocates preference points across traits in a partner
age, field, race, samerace, interests_correlate	Demographics and pair-level similarity
attractive_o, like, decision, decision_o	Partner's ratings and decisions made during or after the date (post-interaction)
match (outcome)	1 if both people said yes to a second date, else 0

Outcome variable: We predict *match*. Matches are a minority class (roughly one in six dates, which we will confirm in exploratory analysis), so accuracy alone would be misleading.

Data characteristics: Many features have missing values (e.g., *guess_prob_liked* is missing in one of the sample rows), and each numeric column also has a binned *d_*-prefixed twin (such as *d_attractive* = "[6-8]") that only re-encodes the same information, so we drop those. Several features are recorded *after* the date (partner ratings, *like*, *decision*, *decision_o*). Since *match* is a direct function of both decisions, using them would be target leakage, and we exclude them from the predictors.

Planned Approach:

Features and cleaning: Our predictors are the five self-ratings (attractiveness, fun, intelligence, sincerity, ambition), plus *expected_num_interested_in_me* and

expected_num_matches as measures of self-confidence about dating success. We will impute missing values (median for numeric features, with a missingness indicator where informative) or drop rows if missingness is rare, and we will inspect distributions and correlations between traits, since self-ratings tend to cluster in the 6–9 range.

Models: We start with logistic regression as an interpretable baseline whose coefficients show the direction and strength of each trait's association with matching. We then compare it against a decision tree and a random forest (Breiman, 2001) to test whether nonlinear effects or trait interactions improve prediction.

Evaluation. Because one person appears in many rows, random row-level splits would let a model see the same participant in both training and test data and inflate performance. We will therefore use grouped k-fold cross-validation by participant. We report ROC-AUC and precision-recall AUC, plus precision, recall, and F1 at a tuned threshold, and compare against a majority-class baseline. Class imbalance is handled with class weights.

Stretch goals:

1. Stepwise feature selection. We will search all 123 columns for the features most predictive of a match, and check how much the five self-ratings add beyond pair-level features such as age gap, shared interests, and same race.
2. The perception gap. We will train the same models on how dates rated each person, and measure how far self-ratings sit from others' ratings. This separates "who you say you are" from "how you are perceived."
3. Split by gender. We will test whether self-perception predicts matches differently for men and women, echoing Fisman et al. (2006).

Relation to prior work:

The dataset was introduced by Fisman et al. (2006), who used regression analysis to study gender differences in mate selection and found, for example, that women place greater weight on intelligence and race than men do. Their focus was on stated versus revealed preferences rather than on predictive modeling. Eastwick and Finkel (2008) found that people's stated preferences for traits like attractiveness and earning potential did not predict their in-the-moment attraction to actual partners, which raises the question of how much self-report can predict at all.

Closest to our project, Joel, Eastwick, and Finkel (2017) applied random forests to a speed-dating dataset with more than 100 self-reported measures. They could predict how much people desire particular partners only modestly, and mostly from relationship-specific (dyadic) variation rather than from individual traits. Our project differs in three ways. First, we restrict predictors to a small, interpretable set of self-ratings, which tests how far a lightweight, easy-to-collect profile can go. Second, our outcome is mutual match rather than an individual's desire ratings. Third, we explicitly guard against leakage and participant-level dependence, which basic analyses of this dataset often overlook. We expect modest predictive power and will report it honestly, since a small effect size is itself an informative finding.

References:

- Breiman, L. (2001). Random forests. *Machine Learning*, *45*(1), 5–32.
- Eastwick, P. W., & Finkel, E. J. (2008). Sex differences in mate preferences revisited: Do people know what they initially desire in a romantic partner? *Journal of Personality and Social Psychology*, *94*(2), 245–264.
- Fisman, R., Iyengar, S. S., Kamenica, E., & Simonson, I. (2006). Gender differences in mate selection: Evidence from a speed dating experiment. *The Quarterly Journal of Economics*, *121*(2), 673–697.
- Joel, S., Eastwick, P. W., & Finkel, E. J. (2017). Is romantic desire predictable? Machine learning applied to initial romantic attraction. *Psychological Science*, *28*(10), 1478–1489.
- Vanschoren, J., van Rijn, J. N., Bischl, B., & Torgo, L. (2013). OpenML: Networked science in machine learning. *SIGKDD Explorations*, *15*(2), 49–60. Dataset: OpenML ID 40536 (SpeedDating).